Example 7.3

A farmer has a field in the shape of a rhombus. The perimeter of the field is 400m and one of its diagonal is 120m. He wants to divide the field into two equal parts to grow two different types of vegetables. Find the area of the field.

Solution

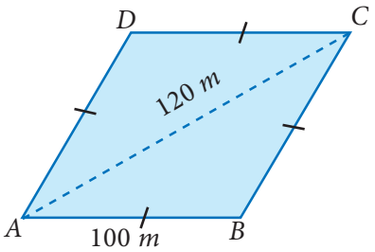


Fig. 7.8

Let $ABCD$ be the rhombus.

Its perimeter = 4 × side = 400 m

Therefore, each side of the rhombus = 100 m

Given the length of the diagonal AC = 120 m

In ΔABC, let a =100 m, b =100 m, c =120 m

$$s = \frac{a+b+c}{2} = \frac{100+100+120}{2} = 160\ m$$

$$\text{Area of } \Delta ABC = \sqrt{160(160-100)(160-100)(160-120)}$$

$$= \sqrt{160\times 60\times 60\times 40}$$

$$= \sqrt{40\times 2\times 2\times 60\times 60\times 40}$$

$$= 40\times 2\times 60 = 4800\,m^2$$

Therefore, Area of the field $ABCD = 2\times$ Area of $\Delta ABC = 2\times 4800 = 9600\ m^2$

Exercise 7.1

1. Using Heron's formula, find the area of a triangle whose sides are

   (i) 10 cm, 24 cm, 26 cm (ii) 1.8m, 8 m, 8.2 m

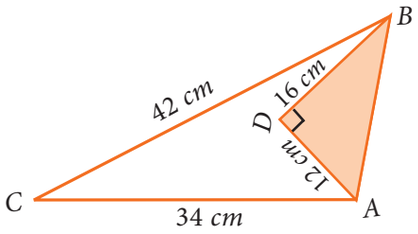
2. The sides of the triangular ground are 22 m, 120 m and 122 m. Find the area and cost of levelling the ground at the rate of ₹ 20 per m^2.

3. The perimeter of a triangular plot is 600 m. If the sides are in the ratio 5:12:13, then find the area of the plot.

4. Find the area of an equilateral triangle whose perimeter is 180 cm.

5. An advertisement board is in the form of an isosceles triangle with perimeter 36m and each of the equal sides are 13 m. Find the cost of painting it at ₹ 17.50 per square metre.

6. Find the area of the unshaded region.



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